SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

Register Number	**192311097**
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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

<u>Descriptive Test</u>

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I <u>*DILLILOKESH D (192311097)*</u>*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

<u>RUBRICS FOR CALCULATION:</u>

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

HOME BASED AUTOMATION SYSTEM

A smart home automation system is a system that automatically controls home devices such as lights, fans, air conditioners, and security cameras based on the surrounding environment and the user's daily activities. The system receives information from different sensors like motion sensors, temperature sensors, and time settings. It then makes decisions to improve comfort, save electricity, and provide better security. Different types of intelligent agents can be used in such a system, and each agent behaves in a different way.

<u>Simple reflex Agent:</u>

A Simple reflex agent works only the according to the current situation. It does not remember the what has happened before and follows fixed rules.

This type of agent follows the most intelligence because it tries to maximize the overall satisfaction instead of only achieving one goal.

So the advantage is :

Balance multiple factors.
Learns user preference.

Limitations:

More difficult to design.
Requires the continuous evaluation of different choices

<u>Model based agent:</u>

A model based agent keeps the current situation and information to the state of home itself. It will remembers the previous sessions and know all the information of last session and it helps to take the descision making system

If we take the example we are entering the house and lights are the not on and it will allows the wheather already the switch is on or off and detects and find the solution for the agent that are currently running and make the decision and evaluates the result do the automation.

The main advantage:

It stores the previous memory of the last session
Reduce the unnessary decision.

Limitations:

More complex than a simple reflex agent

Goal-Based Agent:
In this goal based agent it will take the descision based answer for the question with achieving the specific objective.

EXAMPLE:
If we are maintain the current bill we need to switich of the light and fan to reduce the electricity bills and it has the specific task to save the money and humans missing part it will do by itself .

Advantages:
Focuses on achieving specific goals
Can handle different situations
Better decision-making

Limitations:
The limitation are the needs of planning to switch on the fan or light we need to assign the constraints.

Utility-Based Agent:
A utility-based agent selects the action that provides the highest overall benefit. It compares different options and chooses the one that gives the best balance between comfort, energy saving, and security.

Example
In summer we are using air conditioner for cooling the machine cannot know that we are using the air conditioner with this agent we can find the on which time we can uses the most current bill to less and reliable to the condtion based.

Advantage :

It restores the current and useablity.
It will take the decision on assigned time.

Limitations:
In this agent we want to feed the content to the agent
The automation process will work on the processed agent.

Conclusion: model based , goal based and utility based are effective

Uniform search stratergies:

Uniform Cost Search is a search algorithm that always selects the path with the lowest total cost from the starting point. Instead of searching level by level, UCS checks the path cost and expands the least expensive path first.

Example :

in a maze, some paths may be longer or may have different movement costs. The robot keeps calculating the total cost of every possible path and always chooses the one with the minimum cost.

Advantages

Always finds the optimal path if all movement costs are positive.

Works well when different paths have different costs.

Iterative Deepening Search

Iterative Deepening Search combines the ideas of Depth First Search and Breadth First Search . It starts searching with a small depth limit and increases the limit step by step until the goal is found

Example:

the robot first searches one level deep. If the exit is not found, it searches two levels, then three levels, and so on until it reaches the exit..

Advantages

Uses very little memory.

Can find the shortest path when all movements have equal cost.

Uniform cost search	Iterative Deepening Search
Lowest path cost	Increases depth limit gradually
Find the optimal cost	optimal cost paths are equal
Time complexiety are high	time complexity also high
Space complexity also high	space complexity are low
It uses the different path cost	it an unkonw search depth.